

DUNCAN BARNES

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SOFTWARE ENGINEER

Full-Stack Development | AI/ML & Computer Vision | Cloud-Native Applications

SUMMARY

Software Engineer specializing in AI-driven applications, full-stack development, and scalable cloud-native solutions. Experienced building intelligent systems with TensorFlow, computer vision, and real-time data processing. Skilled in Python, C++, and JavaScript with a strong foundation in secure software development, API design, and CI/CD automation. Proven ability to ship production code, lead cross-functional teams, and deliver measurable impact in fast-paced environments.

Expertise includes

- AI/ML & Deep Learning (TensorFlow)
- Full-Stack Web & Mobile Development
- Cloud Architecture (Azure, GCP, Firebase)
- Security Automation & API Integration

SKILLS

Languages: C++, Python, Java, JavaScript, SQL, HTML/CSS

Frameworks & Libraries: React.js, Flutter, TensorFlow, OpenCV

Systems & Security: Linux (Ubuntu, Kali), Endpoint Security, Threat Detection, Secure Software Development, Cryptography (PKI, Symmetric/Asymmetric Encryption), Incident Response

Cloud & DevOps: Microsoft Azure, Google Cloud Platform, Firebase, GitHub Actions (CI/CD), REST APIs

Tools: Git, GitHub, Jira, Confluence, Prometheus, Microsoft Defender, Vectra

EXPERIENCE

AI Roboto EDU (LIT)

Remote

Software Developer Intern

Apr 2026 – Jul 2026

- Built FixSense, an end-to-end Python pipeline processing 12,000+ car-repair videos
- Ran Whisper transcription, keyframe extraction, and VLM labeling (Qwen3-VL/Claude)
- Indexed labeled actions into a Qdrant vector store for semantic search
- Added AI grading with Claude vision to score output, reporting to Excel/CSV
- Built a marker-pdf manual pipeline and shipped a WordPress site (GCP) + JS app

Fremont Bank

Livermore, CA

Information Security Engineer and Analyst

Oct 2024 – Mar 2026

- Built security automation on the Defender and Vectra APIs, cutting triage time 50%
- Automated threat containment: endpoint mitigation 15 min to under 2 across 500+ systems
- Correlated multi-source telemetry to improve detection and cut false positives
- Reviewed code and validated automation scripts for secure-software practices

IAmI Authentications

Remote

Software Engineering Intern

Aug 2023 – Nov 2023

- Developed secure web application features using React.js and Python, contributing over 500 lines of modular code
- Built and tested dynamic UI components for real-time authentication workflows
- Established and maintained a secure ngrok tunnel for live API testing and webhook simulation
- Conducted over 10 code reviews, enforcing best practices in JavaScript and React development

PROJECTS

Sign Language AR AI Application | HoloLens 2, TensorFlow, Computer Vision

- Developed HoloLens 2 application integrating AI inference models in a constrained hardware environment
- Implemented TensorFlow-based model for gesture classification with focus on performance optimization and real-time data processing

Video-Based Social Media Platform | iOS, Android, Cross-Platform

- Led a team of 5 in designing a video-based platform connecting students to employers; placed 2nd and 3rd in two business plan competitions

Interactive Campus Map | Flutter, Firebase, Google Maps API

- Engineered Flutter-based application enabling users to report campus safety concerns in real time
- Integrated Firebase backend with structured event logging and secure API communication

EDUCATION

California Baptist University

Riverside, CA

Bachelor of Science in Computer Science

Relevant Coursework: Data Structures, Algorithms, Artificial Intelligence, Software Engineering, Operating Systems & Networking, Human-Computer Interaction, Graphics & Visual Computing, Embedded Systems

Leadership: President, National Society of Black Engineers (2023–2024)